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LoRaWAN-based system for multivariate data acquisition in high Andean agriculture

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ABSTRACT

High-Andean agriculture in Peru faces significant challenges from climate change and a scarcity of reliable data for precise decision-making. Remote monitoring in these zones is constrained by a lack of Internet connectivity, particularly in hard-to-reach agricultural areas. This study presents a prototype Internet of Things (IoT) system based on LoRaWAN technology for the collection and storage of multivariate data tailored for agriculture in high-Andean zones. The system integrates sensors for temperature, humidity, solar radiation, wind speed, and wind direction, transmitting the information via long-range, low-power communication to an IoT server for storage and analysis. Field tests, conducted at 3800 m.a.s.l., achieved communication distances of 5 km and 10 km with a 100 % successful transmission rate. The sensor nodes were solar powered, enabling autonomous and continuous operation. The results demonstrate that LoRaWAN can overcome connectivity barriers in high-altitude rural environments, offering a scalable and sustainable solution that strengthens climate resilience and optimizes agricultural practices in high-Andean regions.

1. Introduction

The Long-Range Wide-Area Network (LoRaWAN) protocol has become one of the most effective solutions in the Internet of Things (IoT) domain [1]. This advanced technology enables the connection of multiple devices to the internet, playing a fundamental role as an enabler for IoT systems [2] LoRaWAN operates in unlicensed frequency bands and employs the Chirp Spread Spectrum (CSS) technique, which is characterized by the use of wide-spectrum linear frequency chirps. This technique is foundational for Low-Power Wide-Area Networks (LPWANs) [3,4]. Furthermore, its low energy consumption and resilience to electromagnetic spectrum interference make LoRaWAN an ideal technology for applications in adverse environments [5]. LoRa radio transceivers operate at frequencies of 433 MHz and 868 MHz, with transmission powers of up to +10 dBm (10 mW) and +14 dBm (25 mW), respectively, allowing for flexibility across various regions [6]. Devices connected to LoRaWAN are referred to as low-power wireless IoT devices and can be classified according to the range of the networks in which they operate:

- Local Area Networks (LANs) and Personal Area Networks (PANs), which encompass short-range, low-power communication for devices with a range below 1000 m. These networks include technologies such as Bluetooth, IEEE 802.15.4, and IEEE P802.1ah.
- Wide Area Networks (WANs) are characterized by a communication range greater than 1000 m, serving low-energy consumption devices.

Agriculture faces complex challenges stemming from climate change, variations in soil quality, and the incidence of pests, frosts, and droughts [7,8]. The characterization of agricultural processes requires reliable data to assess spatial and temporal variability, which entails the consideration of multiple interdependent variables. In this context, advancements in data science and analytical tools present significant opportunities to integrate and analyze agricultural data cost-effectively and automatically, in real-time and with high precision [9]. However, the implementation of Big Data in smart agriculture remains limited by a scarcity of data in regions where Andean agriculture is prevalent, which underscores the need to apply new technologies and to continue advancing this field.

In addition to these general challenges, climate change is generating alarming impacts on water and climatic resources across various regions, particularly affecting high-Andean zones. These areas, situated in the Puna grasslands of the Andes at altitudes exceeding 3500 m.a.s.l.,

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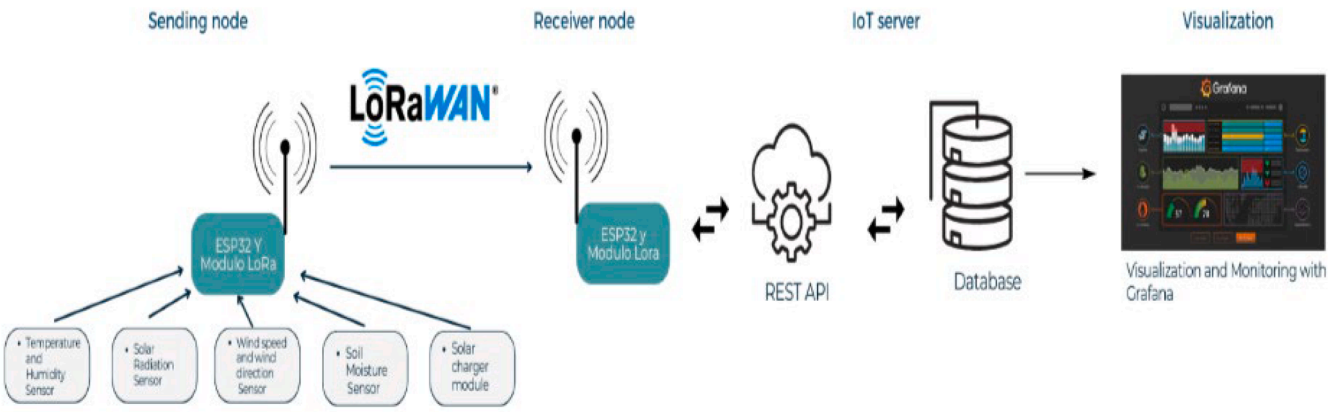


Fig. 1. System architecture of the proposed LoRaWAN prototype.

encompass territories in countries such as Colombia, Ecuador, Peru, and Bolivia [10]. Due to their ecological fragility and the limited socioeconomic capacity of their inhabitants, these regions are highly vulnerable. They face significant risks, including biodiversity loss and ecosystem degradation, with extremely slow recovery rates following disturbances [11]. However, despite their environmental importance, high-Andean zones have received limited attention in environmental policies, in contrast to tropical forests, which are prioritized for conservation [12].

In this context, the objective of the present research is to develop a prototype based on a LoRaWAN network for the collection and storage of multivariate data in high-Andean agriculture. This prototype is designed to operate efficiently in remote environments, enabling the real-time collection and storage of multiple agricultural parameters, such as soil moisture, temperature, wind speed and direction, solar radiation, and atmospheric pressure. The graphical representation of the collected data will facilitate its interpretation by farmers, allowing them to make informed decisions and optimize their agricultural practices. Thus, this research seeks to provide a technological solution tailored to the specific needs and limitations of high-Andean zones, promoting the use of IoT and Big Data to strengthen sustainable development in these regions.

2. Related work

This section presents a review of state-of-the-art monitoring systems for precision agriculture, with a focus on the wireless communication technologies that enable in-field data collection. The analysis centers on identifying existing solutions, their applications, and, crucially, their limitations, in order to contextualize and highlight the research gap that the present work addresses within the unique ecosystem of the high-Andean zones.

The integration of advanced technologies into agriculture is a rapidly expanding field of research. Recent reviews [5,13] highlight how the Internet of Things (IoT) is fundamental for monitoring key variables, although they also underscore significant barriers to adoption, such as high initial costs and a lack of connectivity infrastructure. The potential of these technologies is realized in applications that combine data collection with predictive analysis. For instance, [14,15] implemented a Zigbee sensor network to predict apple scab, while European projects analyzed in [16,17] use machine learning to optimize crop management. Nevertheless, an implicit limitation of these studies is their reliance on a relatively accessible connectivity infrastructure.

In rural environments with complex topography or without cellular coverage, the choice of communication technology is a critical factor. For this project, various LPWAN technologies were considered, including NB-IoT and LoRaWAN. While NB-IoT offers higher data transfer rates and benefits from existing cellular infrastructure, it presents critical limitations for the high-Andean context, primarily its dependence on mobile operator coverage, which is non-existent in many of the agricultural zones of Puno. In contrast, LoRaWAN was chosen for its independence from infrastructure, its low power consumption, and its lower operational cost by not requiring subscriptions to telecommunication services, making it a more viable and sustainable solution.

The effectiveness of LoRaWAN has been demonstrated across a diverse landscape of applications, including industrial energy control [18,19] and smart home automation [1]. Within environmental monitoring, many works rely on third-party cloud services for data management. For example, systems for water quality surveillance [20,21] and flood alerts [2,22] frequently utilize platforms like The Things Network (TTN) and ThingSpeak. Similarly, studies on farms, such as those by [23,24], employ ThingSpeak or Blynk for visualization. Unlike these approaches, one of the contributions of our work is the development of a custom IoT server with its own REST API. This architecture guarantees complete control over data management, security, and customization, a fundamental aspect for creating a robust and data-sovereign solution in the region.

Despite its success in these applications, most have been conducted under moderate environmental conditions and focus on the collection of low-payload data. It is acknowledged that transmitting data at higher rates, such as images, remains a technical challenge for LoRaWAN. Cutting-edge research, such as that explored in [25], is developing staged processing systems based on deep learning to optimize the selective transmission of images. This justifies our study's focus on the reliable collection of sensor data, which constitutes the most fundamental and unmet need in the studied region.

Although the existing literature on LoRaWAN is extensive, a notable gap persists in its validation within extreme geo-climatic ecosystems. Agriculture in the Peruvian Andes, crucial for local food security, faces unique challenges documented by the Climandes project [11]. This work highlights the imperative need for high-resolution local data to counteract the low predictive capacity of climate models in the region. Therefore, our research directly addresses this gap by presenting the in-field validation of a system designed to generate the fundamental data that is currently non-existent, thereby strengthening agricultural resilience in the challenging context of the high-Andean zones.

3. Materials and methods

3.1. Component selection

The components for the LoRaWAN system were selected considering the characteristic environmental factors of the high-Andean zones, such as extreme temperature variations, limited solar radiation, and relative humidity. The selection process prioritized the cost-benefit ratio, ensuring that the sensors were reliable, cost-effective, and had a minimal margin of error in measuring the study variables. Thus, the total cost

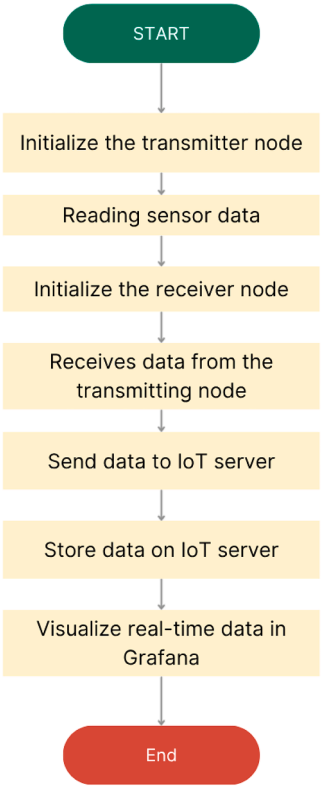


Fig. 2. Flowchart of the data acquisition and storage process.

of the system amounts to approximately $320.00 USD.

The hardware components comprise an ambient temperature and humidity sensor (CWT-TH04S, ComWinTop), a soil temperature sensor (DS18B20), a solar radiation sensor (sn-300al-ra-n01), an anemometer (CWT-SWS-C-RS485), a wind direction sensor (zts-3000-fxjt-n01), a microcontroller (ESP32 WROOM 32 CP2102), a LoRa module (E220–900T22D-V1.2), an 8000 mAh 18,650 battery, a solar panel (Cafini 12 V 37 W), a charge controller (Solar MPPT 30A), and a voltage converter (Mini560 5 V, 12 V). The software components include the Arduino IDE 2.3, the Visual Studio Code editor version 1.100.3 for IoT server development, and the free online version of SketchUp for 3D designs.

3.2. *Proposed architecture of the LoRaWAN-based prototype*

The prototype's architecture is based on a wireless sensor network that utilizes LoRaWAN technology, integrated within the context of the Internet of Things (IoT), for data transmission in hard-to-reach areas. In this framework, ESP32 microcontrollers were employed as the central processing unit to integrate industrial-grade sensors via the Modbus communication protocol. This protocol facilitates the collection of critical environmental parameters, such as air and soil temperature, ambient humidity, soil moisture, wind direction and speed, solar

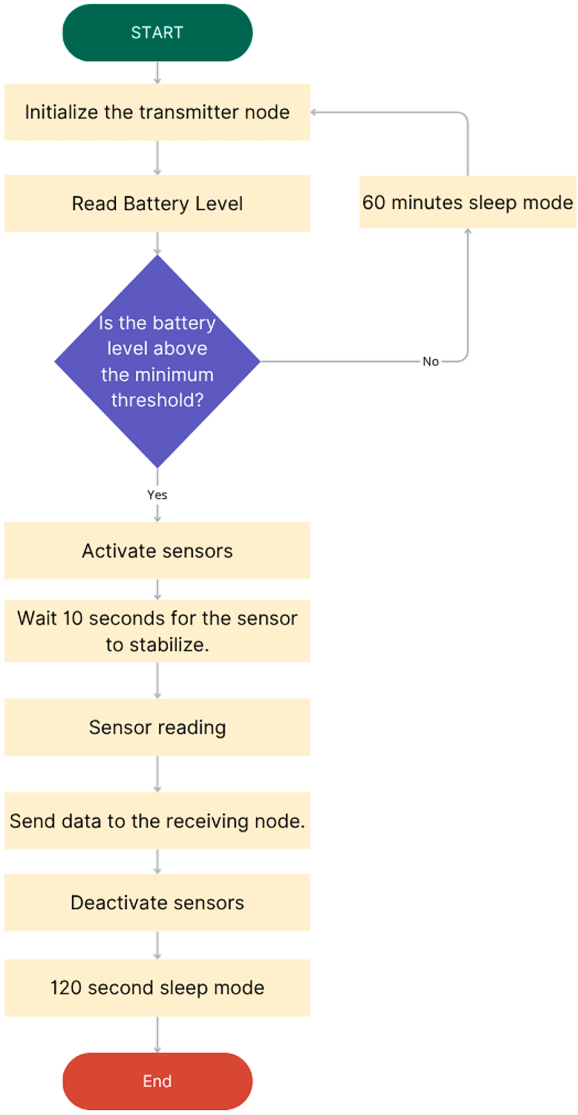


Fig. 4. Flowchart of the transmitter node.

radiation, and atmospheric pressure, obtained from the aforementioned sensors. Fig. 1 presents the block diagram of the transmitter and receiver nodes, integrated with the ESP32 microcontroller and the LoRa module for data storage and visualization. Additionally, it shows the interconnections between the components and describes the overall architecture of the developed prototype (Fig. 1)

A simplified flowchart in (Fig. 2) provides an overall perspective of the data acquisition and processing system. This system is structured around two main nodes: a transmitter node and a receiver node, both integrated with ESP32 microcontrollers and LoRa modules for wireless communication.

The process begins with the initialization of the transmitter node,

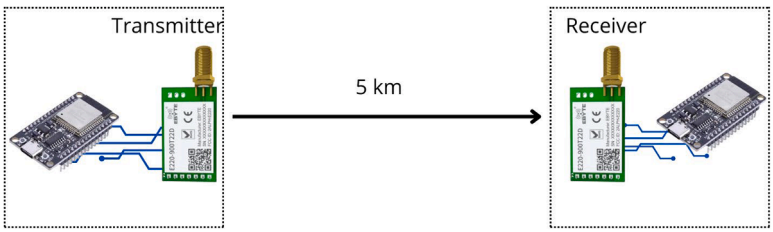


Fig. 3. The transmitter and receiver node devices.

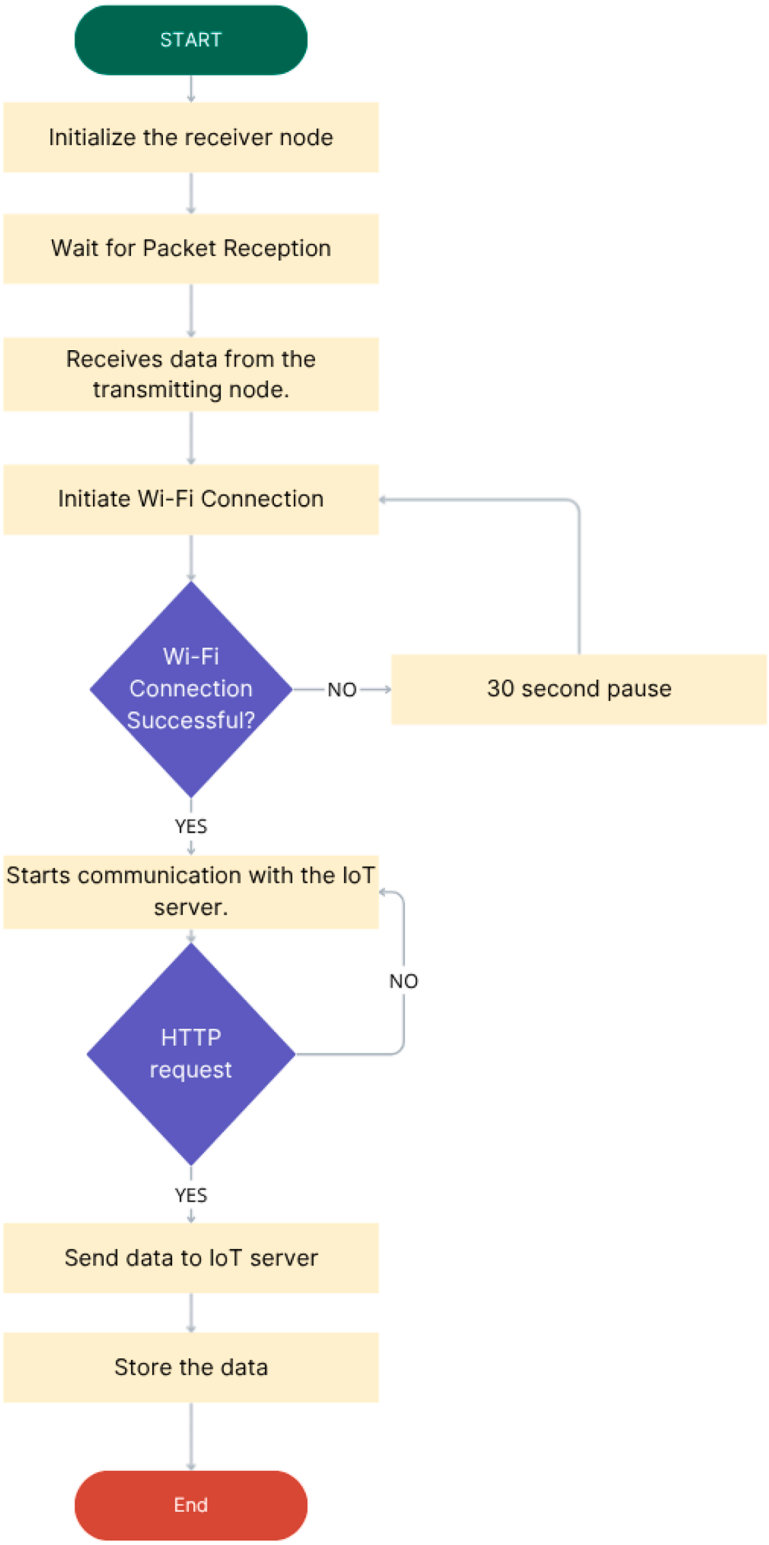


Fig. 5. Flowchart of the receiver node.

whose function is to read data from the connected sensors. Subsequently, the receiver node is initialized. Once operational, the receiver node is prepared to receive data wirelessly sent by the transmitter node via the LoRa module. Following successful reception, the receiver node utilizes a custom REST API to send the data to the IoT server, where it is securely stored. Finally, the stored data is visualized in real-time using the Grafana platform, concluding the general operational flow of the system.

3.3. LoRaWAN prototype design

Fig. 3 presents the design of the LoRa-based wireless communication prototype, which consists of a transmitting and a receiving station, with operational ranges of 5 km and 10 km under direct line-of-sight (LoS)

conditions. The prototype was implemented in high-Andean zones where the stations maintain a visual field free of obstacles, thereby reducing the electromagnetic interference common in urban environments. The configuration of the modules was performed using EBYTE's software, establishing parameters such as the transmission channel, access key, and output power for subsequent serial communication with the ESP32 microcontroller. Both the transmitter and receiver modules were configured with identical parameters to ensure compatibility and correct operation. Furthermore, the frequency bands permitted in the region were considered, opting for 915 MHz in accordance with local regulations.

The data transmission mechanism of the transmitter node is detailed in the flowchart presented in (Fig. 4). This node is powered by a solar panel that charges a rechargeable 18,650 LiPo battery, which energizes the ESP32 microcontroller. The operational process begins with the initialization of the transmitter node, followed by a battery level reading. This battery level information is also sent and stored on the IoT server for diagnostic purposes, allowing for an understanding of the node's energetic history. At this point, the system evaluates whether the charge level is above a predefined minimum threshold. If the battery is below this threshold, the node enters a low-power sleep mode for 60 min to preserve the remaining energy. After this period, the battery verification cycle restarts, seeking the optimal condition for operation. Conversely, if the battery level is adequate, the node proceeds by activating its multivariable sensors via a relay. Subsequently, it waits for a 10-s interval for sensor stabilization. Once the sensor data reading is completed, these data are sent to the receiver node through the LoRa module. Upon completion of the transmission, the sensors are deactivated via the relay to optimize consumption, and the node enters a 120-s low-power sleep mode before restarting the operational cycle with a new battery level reading. This design optimizes energy efficiency and ensures data continuity and integrity in the high-Andean zones.

The operational flowchart of the receiver node, illustrated in (Fig. 5), is designed for the reception and management of signals sent by the transmitter node. The process begins with the initialization of the receiver node, which prepares it for its main function. Subsequently, the node enters an active wait state for LoRa packets from the transmitter node.

Once a LoRa data packet is detected and received, the node proceeds to initiate a Wi-Fi connection. At this point, the system evaluates Wi-Fi connectivity. If the connection is not established successfully, the node implements a 30-s pause before reattempting the connection, establishing a recovery cycle. If the Wi-Fi connection is successful, the node advances to initiate communication with the IoT server via an HTTP request. The success of this request is internally validated. If the HTTP request is unsuccessful, the node reattempts communication with the IoT server. Conversely, if the HTTP request is successful, the data are sent to the IoT server and are stored in the database.

3.4. Software architecture

The software development is based on a client-server architecture, which integrates the system components using specific programming languages for distinct functions. First, *C++* was utilized for programming the ESP32 microcontrollers, enabling communication with the multivariate sensors and wireless data transmission via LoRaWAN and Wi-Fi. Second, Laravel was employed for the development of the server's backend, which is responsible for the secure processing and storage of data. Finally, this architecture enables data transmission to the cloud via a REST API, ensuring seamless and reliable communication between the nodes and the IoT server.

3.5. IoT server

Cloud services have transformed data management in IoT environments, enabling efficient storage and remote access from anywhere in the world. Nevertheless, in this study, a custom IoT server was developed with the objective of ensuring complete control over the system's data and operations.

This approach allows for the complete customization of the system's integration according to the specific needs of the project. In contrast to cloud solutions, which impose limitations in terms of configuration and functionalities, a custom IoT server offers greater flexibility, optimizing both performance and the implemented security measures.

3.6. Data visualization

For real-time data visualization, Grafana was utilized, a tool that specializes in the graphical representation of data and metrics. This platform enables the monitoring of the proper storage of information in the database and the rapid detection of any anomalies or data gaps through dynamic graphs. Additionally, Grafana incorporates API services for sending notifications to users via platforms such as WhatsApp or Telegram, alerting them to critical events within the system.

4. Results and discussion

4.1. Challenges and communication technologies in high-andean agriculture

An evaluation of currently available communication technologies for application in high-Andean agriculture was conducted, with a focus on GSM/2G/3G/4 G and Wi-Fi networks. While GSM/2G/3G/4 G networks provide broader coverage, they present significant challenges, including high implementation costs, the need for ongoing maintenance, and limited connectivity in high-Andean zones due to the mountainous terrain. In these regions, GSM signal reception is often restricted to elevated areas where a connection is possible. However, agricultural activities are typically carried out on mountain slopes to mitigate abrupt temperature changes, thereby hindering access to these networks.

On the other hand, Wi-Fi is a commonly used technology that offers robust broadband but has a limited range of 100 m in open areas. Although antennas exist that can extend this range, their high costs and the lack of a constant power supply in high-Andean zones hinder their implementation. Additionally, these regions are prone to frequent electrical storms, which demands special care in the installation and maintenance of communication equipment.

4.2. Functional evaluation and prototype design optimization

Functionality tests of the prototype were conducted, evaluating both the hardware and the software. During this phase, several errors related to the electronic circuit's power supply and the microcontroller's firmware were identified and corrected. The system employs a 12 V, 36 W solar panel, sensors with a supply voltage range of 7 V to 24 V, and a 5 Ah lithium-ion battery based on 18,650 cells. When operating in the Andean zone, at approximately 4000 m.a.s.l., temperatures frequently drop below 0 °C and atmospheric pressure is lower. These conditions increase the internal resistance of the cells and reduce their effective capacity, leading to an accelerated discharge that compromises the system's autonomy.

Operation in the Andean zone ($\approx$ 4000 m.a.s.l.) poses additional challenges, as low temperatures (frequently < 0 °C) and lower atmospheric pressure increase the internal resistance of the cells, reduce their effective capacity, and accelerate discharge, thereby compromising autonomy. To mitigate these effects, a sleep mode was incorporated into the microcontroller. Thus, the transmitter node transmits data approximately every 110 s and remains in a sleep state for the rest of the time. Additionally, a relay was integrated to disconnect the power supply to the sensors during the sleep periods, minimizing energy losses. Prior to each reading, a stabilization interval (warm-up) is executed to ensure the quality of the measurements before transmission.

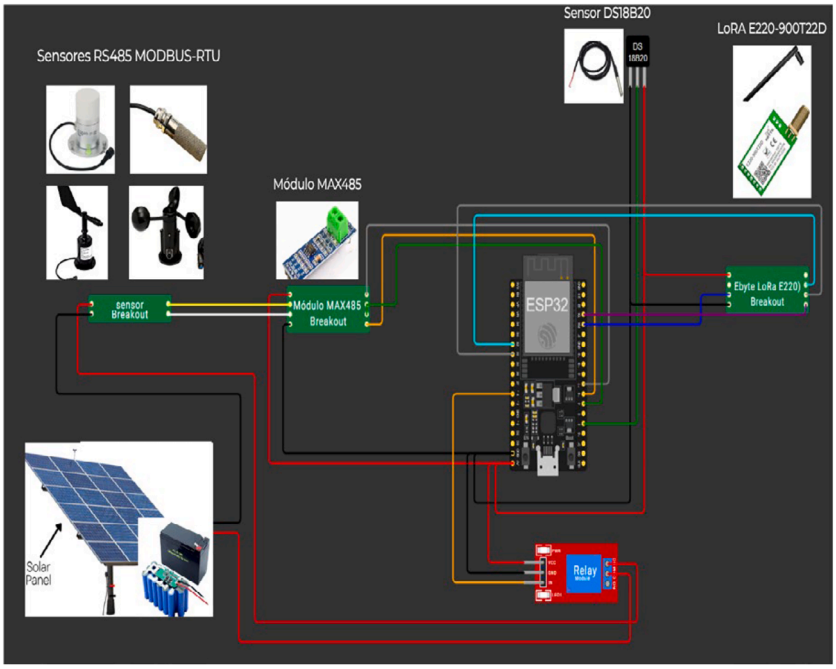


Fig. 6. Schematic diagram of the transmitter node.

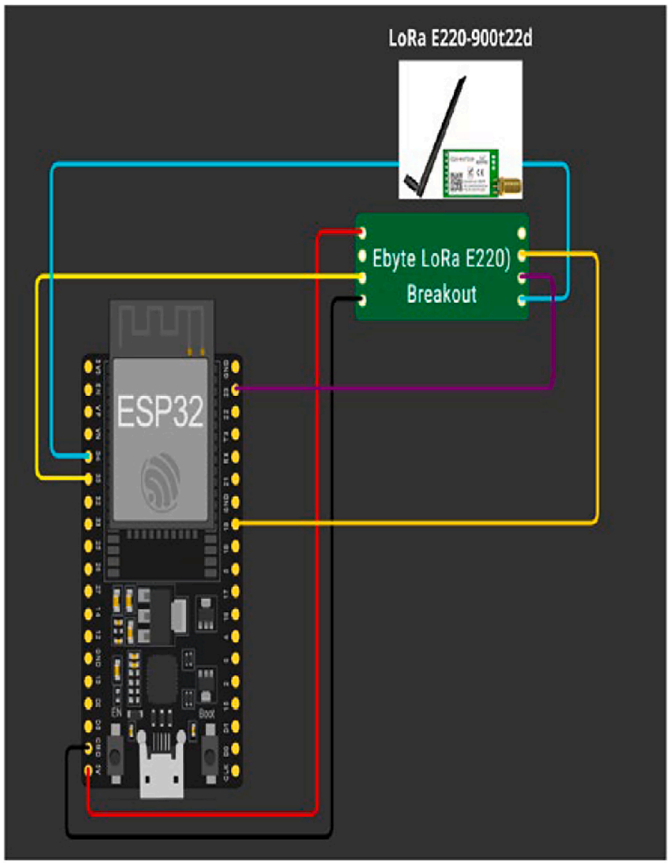


Fig. 7. Schematic diagram of the receiver node.

The transmitter node connection diagram is depicted in (Fig. 6), whereas the receiver node connection diagram is shown in (Fig. 7).

Individual tests of each sensor were performed by compiling their respective code in the Arduino IDE in order to verify correct readings

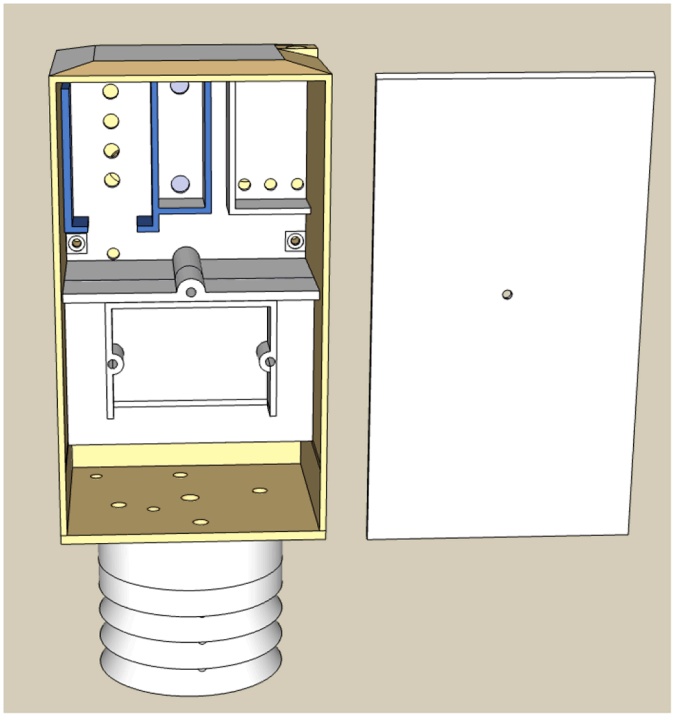


Fig. 8. 3D design of the transmitter node enclosure.

and isolate potential conflicts prior to full integration. During this stage, six sensors were employed, and their output was monitored via the Arduino Serial Monitor. Additionally, the use of color-coded cables facilitated the labeling and connection of each device. To efficiently detect and diagnose faults, debugging routines were incorporated that issue specific messages if any sensor fails to respond.

Once the sensors were validated, the complete system was assembled, including the LoRa module in the transmitter node, and data transmission to the receiver node was initiated. This node decodes the

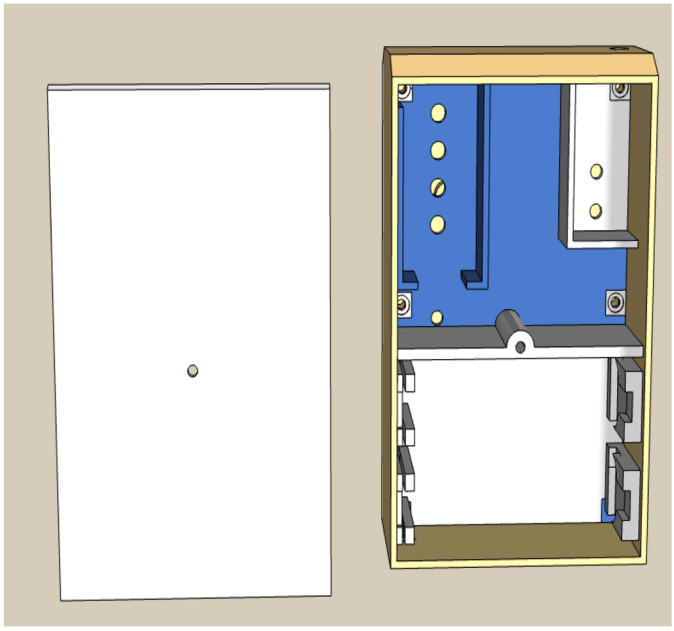


Fig. 9. 3D design of the receiver node enclosure.



Fig. 10. Assembled prototype of the receiver node.



Fig. 11. Assembled prototype of the emitter node.

received frames, processes the information, and, after verifying Wi-Fi connectivity and authentication via the Arduino code, transmits the data to the IoT platform or custom server for storage. This comprehensive evaluation confirms the correct operation of the sensors and LoRa modules in both nodes, as well as the reliability of the data flow to the cloud.

4.3. Design and implementation of the lorawan architecture

Prior to physical deployment, protective enclosures were designed in SketchUp, subsequently generating the STL files for their fabrication via 3D printing. PETG filament was used, selected for its mechanical strength, thermal stability, and low moisture absorption—properties that make it ideal for in-field applications. The result was a set of parts with a precise fit for the electronic circuitry, capable of effectively protecting the components from rain and adverse environmental conditions, thus ensuring the continuous operation of the system.

The enclosure for the transmitter node is presented in (Fig. 8), while the design for the receiver node is shown in (Fig. 9); both were optimized to provide structural robustness and facilitate the easy integration of the electronic modules.

To implement the LoRaWAN architecture, two representative locations in the high-Andean zones of Puno, Peru, were selected: Muñani (4000 m.a.s.l.) and Juliaca (3800 m.a.s.l.). In Muñani, the transmitter node was equipped solely with the humidity and temperature sensor and was installed at an elevated point with a direct Line-of-Sight (LoS) to the receiver, separated by a distance of 10 km. An initial test transmission at this distance confirmed the viability of the link.

To implement the LoRaWAN architecture, two representative locations in the high-Andean zones of Puno, Peru, were selected: Muñani (≈



Fig. 12. Final assembled prototype of the emitter node.

4000 m.a.s.l.) and Juliaca (≈ 3800 m.a.s.l.). In Muñani, the transmitter prototype (emitter node) was equipped with a humidity and temperature sensor and installed at an elevated point with a direct line-of-sight to the receiver prototype (receiver node), separated by approximately 10 km. An initial test transmission at this distance confirmed the viability of the link. The receiver prototype (receiver node) is shown in (Fig. 10), while the transmitter prototype (emitter node) is presented in (Fig. 11), both of which were designed and configured to operate in the high-Andean environment.

In Juliaca, located in a more extensive agricultural area, the emitter node was integrated with all six of the system's sensors. Following coverage verification, the environmental variables were automatically collected and transmitted every two minutes, thereby validating the reliability of the link and the platform's capacity to operate in both environments.

The second implementation phase was conducted in a more extensive, representative agricultural area in Juliaca. In this environment, the emitter node was deployed, with its final design integrating the full set of sensors for multivariate monitoring, as shown in (Fig. 12). Complementing this, the receiver node was installed (Fig. 13), which was responsible for receiving the data and interfacing with the storage platform.

The deployment at this location served to confirm the viability of the proposed architecture in a real-world field scenario, ensuring both the coverage and the proper functioning of the physical components.

4.4. LoRaWAN network deployment in high-Andean zones

To evaluate the effectiveness and range of LoRaWAN technology in the high-Andean context, deployments were conducted in two representative scenarios within the Puno region.

The first deployment was carried out in the district of Muñani, a rural



Fig. 13. Final assembled prototype of the receiver node.

environment characterized by its open topography. In this scenario, a stable communication link was successfully validated over a distance of 10 km with a direct Line-of-Sight (LoS), the geographical location and range of which are detailed in (Fig. 14). Subsequently, the test was replicated in a rural area of Juliaca, an agricultural environment with a higher density of obstacles such as scattered dwellings. The map of this second deployment is presented in (Fig. 15), where a communication link was established over a distance of 5 km..

In both deployments, the data captured by the transmitter node was consistently sent to the receiver node and, from there, to the cloud for storage on the IoT server. This process demonstrated the architecture's capacity to ensure a centralized and efficient management of the information collected under the different topographical conditions of the



Fig. 14. Geographic map of Muñani.



Fig. 15. Geographic map of Juliaca.

region.

4.5. Data collection, storage, and system usability evaluation

The successful reception of data transmitted from the sensor nodes was verified directly in the IoT server's database. Each received data packet was automatically processed and stored, thereby generating a structured record for subsequent querying and analysis.

A screenshot of the database table is presented in (Fig. 16), where the successful storage of the information is evidenced. Each row represents a unique measurement, and the key collected fields can be observed in the columns. These include the values for the primary environmental variables such as temperatura, humedad, presion, suelotemp1, vientokmh, and Radiacion0. Of equal importance, the fechaTReg field records the exact timestamp of each entry, which allowed for the validation of the programmed transmission frequency and served as a tool for detecting transmission incidents.

4.6. Functional validation of the custom iot server

The use of a custom IoT server, developed in this project, demonstrated robust performance by providing complete control over data storage and connection management. Its client-server architecture, based on a REST API, enabled seamless and secure communication between the nodes and the server using standard HTTP methods, such as GET and POST. This design supported multiple nodes distributed across diverse geographical zones, ensuring stability even with a high volume of concurrent requests. To avoid conflicts in data management, dedicated tables were implemented for each zone within the main database, optimizing the handling of the collected information. Additionally, the Grafana platform was used on a laptop to monitor the values obtained from the transmitter node. The multivariate data registered in the database provided valuable information to assess the system's proper functionality and to detect potential anomalies in real-time.

In order to formally describe the periodic behavior of the LoRa nodes, a mathematical model was established to characterize the

	id_temperatura	ubicacion_id	Temperatura	Humedad	FechaTReg	Presion	Altitud	Suelotemp1	Vientoadc	Voltaje	Vientokmh	Viento90	Viento91	Radiacion0
Editar Copiar Borrar	65001	1	-2.40	56.40	2025-08-17 06:14:28	64662.00	3631.81	4.19	0.00	0.00	0.00	4	180	22
Editar Copiar Borrar	65002	1	-2.40	53.30	2025-08-17 06:16:19	64665.00	3631.57	4.19	0.00	0.00	0.00	4	180	23
Editar Copiar Borrar	65003	1	-2.60	53.30	2025-08-17 06:18:09	64664.00	3631.69	4.19	0.00	0.00	0.00	4	180	24
Editar Copiar Borrar	65004	1	-2.60	54.00	2025-08-17 06:20:00	64665.00	3631.69	4.19	0.00	0.00	0.00	4	180	26
Editar Copiar Borrar	65005	1	-2.40	53.90	2025-08-17 06:21:50	64663.00	3631.81	4.12	0.00	0.00	0.00	4	180	27
Editar Copiar Borrar	65006	1	-2.10	53.10	2025-08-17 06:23:41	64668.00	3631.57	4.12	0.00	0.00	0.00	4	180	28
Editar Copiar Borrar	65007	1	-1.90	53.00	2025-08-17 06:25:31	64670.00	3631.45	4.12	0.00	0.00	0.00	4	180	30
Editar Copiar Borrar	65008	1	-1.80	53.10	2025-08-17 06:27:22	64669.00	3630.97	4.12	5.00	0.00	0.00	4	180	32
Editar Copiar Borrar	65009	1	-1.60	52.70	2025-08-17 06:29:12	64672.00	3631.33	4.12	0.00	0.00	0.00	4	180	35
Editar Copiar Borrar	65010	1	-1.40	52.30	2025-08-17 06:31:03	64676.00	3630.85	4.12	0.00	0.00	0.00	4	180	37
Editar Copiar Borrar	65011	1	-1.10	51.60	2025-08-17 06:32:54	64674.00	3630.61	4.12	27.00	0.02	0.00	4	180	39
Editar Copiar Borrar	65012	1	-0.70	51.10	2025-08-17 06:34:44	64675.00	3630.25	4.12	0.00	0.00	0.00	4	180	42
Editar Copiar Borrar	65013	1	-0.30	50.60	2025-08-17 06:36:34	64678.00	3629.17	4.12	0.00	0.00	0.00	4	180	46
Editar Copiar Borrar	65014	1	0.20	49.90	2025-08-17 06:38:25	64688.00	3629.17	4.12	0.00	0.00	0.00	4	180	50
Editar Copiar Borrar	65015	1	0.70	48.70	2025-08-17 06:40:15	64684.00	3629.05	4.19	0.00	0.00	0.00	4	180	55
Editar Copiar Borrar	65016	1	1.10	47.80	2025-08-17 06:42:06	64688.00	3629.29	4.19	0.00	0.00	0.00	4	180	60
Editar Copiar Borrar	65017	1	1.40	47.10	2025-08-17 06:43:56	64699.00	3628.69	4.19	0.00	0.00	0.00	4	180	65
Editar Copiar Borrar	65018	1	1.70	46.40	2025-08-17 06:45:48	64689.00	3628.69	4.19	0.00	0.00	0.00	4	180	72
Editar Copiar Borrar	65019	1	2.20	45.60	2025-08-17 06:47:37	64699.00	3627.74	4.25	0.00	0.00	0.00	4	180	78

Fig. 16. The MySQL database.

operational dynamics of the node. Each operational cycle of the node is composed of two phases: (i) the warm-up phase, where the sensors remain active for a fixed time interval until stable conditions are reached, and (ii) the sleep phase, designed to minimize energy consumption before the next transmission.

To verify that the server stores the records with the planned cadence, the timestamps $T(n)$ (column FechaTReg) associated with each received packet were extracted from the MySQL database (Fig. 16). From these timestamps, the elementary interval between consecutive arrivals was defined as:

$$\Delta t(n) = T(n+1) - T(n) \tag{1}$$

The metric (1) quantifies the regularity (jitter) observed in the server data and constitutes the basis to estimate the operational period.

The experimental period, $t_{periodo}$ was calculated as the average of these elementary intervals, thereby attenuating minor variations caused by network latency or the communication stack:

$$t_{periodo} = \frac{1}{N-1} \sum_{n=0}^{N-2} (T(n+1) - T(n)) \tag{2}$$

When applied to the collected data shown in Fig. 16, this estimation method attenuates minor variations caused by network latency or the communication stack, yielding a stable period of approximately 110 s (Eq. 2).

Knowing the duration of the warm-up phase, t_{warmup} the duration of the sleep phase, t_{sleep} can be determined:

$$t_{sleep} = t_{periodo} - t_{warmup} \tag{3}$$

The resulting value, $t_{sleep} = 95$ s, demonstrates that the dominant fraction of the operational cycle occurs in a low-power state, contributing to significant energy savings.

Definition of Variables

- $T(n)$: Timestamp of the n-th record in the server (MySQL table).
- $\Delta t(n)$: Interval between consecutive records (Eq. 1).
- N: Total number of analyzed records (rows of the fragment shown in Fig. 16).
- $t_{periodo}$: Estimated experimental period (Eq. 2).
- t_{warmup}: Warm-up time for sensor stabilization (fixed in the design; 15 s).
- t_{sleep}: Time spent in the low-power state per cycle (Eq. 3).

4.7. Multivariate data monitoring and visualization

For monitoring, the open-source data visualization platform Grafana was employed. Leveraging its flexibility to connect to various relational and non-relational databases via native drivers and SQL queries, custom dashboards were designed to graphically represent each variable measured by the six sensors in real-time, with updates every two minutes. Grafana facilitates the verification of correct readings from the emitter node; the detection of data gaps in the graphs allows for the identification of transmission failures or inoperative sensors. Optionally,



Fig. 17. Temperature and humidity plot.

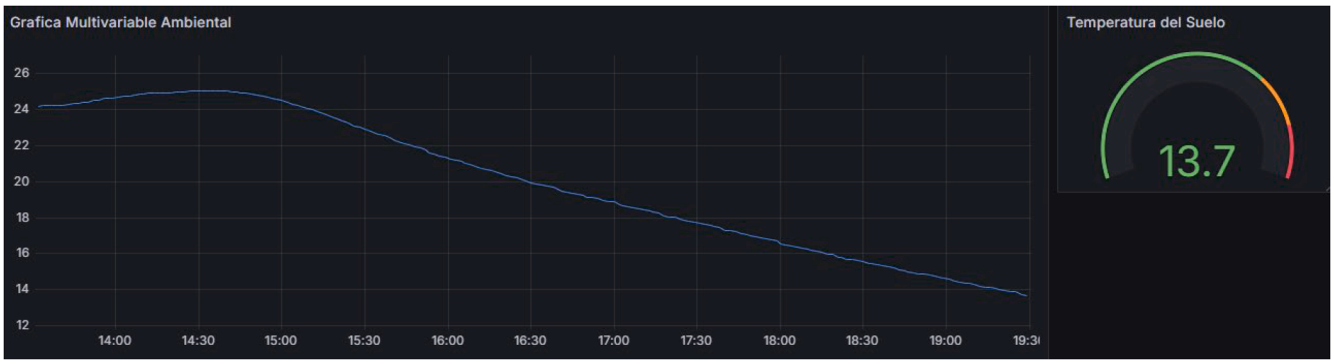


Fig. 18. Diagram of soil temperature.

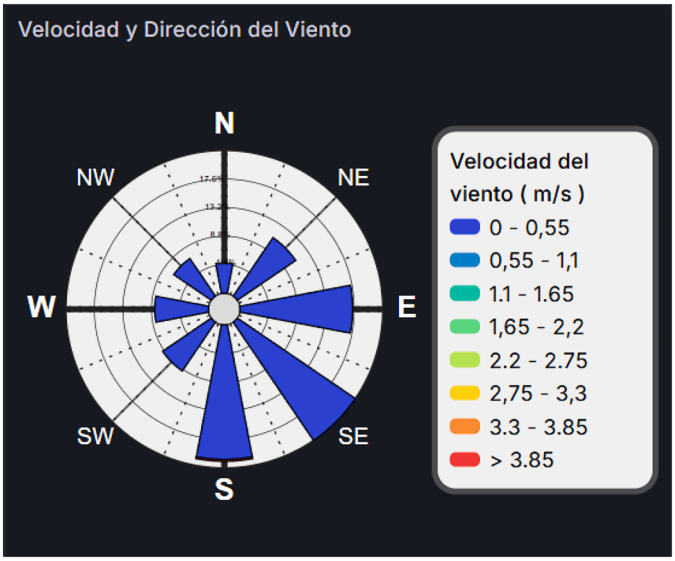


Fig. 19. Wind rose diagram of wind speed and direction.

alerts were also configured to notify users of data update failures.

The dashboard displays the following variables:

The dynamics of ambient temperature and relative humidity are illustrated in (Fig. 17). A clear inverse correlation is observed, which is characteristic of the diurnal cycle. The temperature (green line) begins the period at around 22 °C and steadily decreases to a minimum of approximately 5 °C at the end of the recording period. Conversely, the humidity (yellow line), after remaining at low values (~12 %), begins a marked ascent from 17:00 onwards, exceeding 30 % towards 19:30, coinciding with the drop in temperature at dusk.

Soil Temperature (measured at 5 cm depth)

The soil temperature at a 5 cm depth (Fig. 18) exhibits greater thermal inertia. It reaches its maximum of 25 °C at approximately 14:30, one hour after the start of the recording period, and then follows a smooth and sustained decline to 13.7 °C. This attenuated and delayed behavior is fundamental to understanding the conditions at the crop root level.

The analysis of wind speed and direction, presented in the wind rose diagram (Fig. 19), reveals clear prevailing patterns. The primary wind direction is from the East (E) and Southeast (SE). While the most frequent speeds are low (0 - 0.55 m/s), significant gusts reaching up to 3.3 m/s are recorded, originating primarily from these dominant directions.

The solar radiation profile, detailed in (Fig. 20), is consistent with that of a clear afternoon in a high-altitude region. The recording begins with very high values, close to 1400 W/m2, which progressively decrease until reaching zero at approximately 17:30, marking sunset.

The monitoring of atmospheric pressure is shown in (Fig. 21). The recorded absolute value, around 64,454 Pa (644.5 hPa), is consistent with the altitude of the Puno region (~3800 m.a.s.l.) and validates the sensor's proper calibration in this environment. The plot illustrates the



Fig. 21. Atmospheric pressure plot.

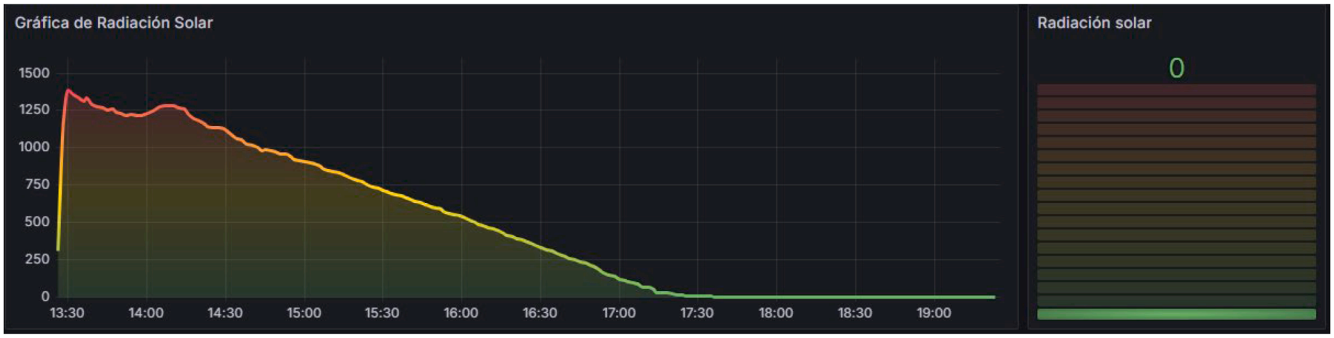


Fig. 20. Solar radiation plot.

natural barometric fluctuations over the six-hour period.

Beyond real-time monitoring, the platform demonstrated its potential for advanced functionalities. Grafana's capability to generate historical reports, which are essential for analyzing seasonal climate trends, was verified, as was its capacity to configure alert thresholds. This latter feature is fundamental to evolving the system into a precision agriculture tool.

5. Conclusions

The objectives of this research were successfully met through the proposal, development, and implementation of a prototype based on LoRaWAN wireless technology. This system enables the real-time collection of multivariate data in the high-Andean zones of the Puno region, Peru, where agriculture is a fundamental activity.

The prototype was developed and implemented in the high-altitude region of Puno, Peru. It proved effective in long-distance data transmission, utilizing LoRa and LoRaWAN modules and low-power microcontrollers. Given the absence of a power grid in these areas, the transmitter node was designed to operate on renewable solar energy, making it an ideal and sustainable solution for the high-Andean context. Initial tests validated its functionality at 10 km, and additional tests in more densely populated agricultural areas further confirmed its performance. The system enabled internet communication, cloud data storage, and visualization on web and mobile platforms, facilitating continuous and accessible monitoring.

These results are promising, as the implementation of this system can significantly improve the decision-making of farmers and local authorities by providing real-time information on the behavior of environmental variables.

It is important to acknowledge that the system's performance may be affected by adverse climatic factors characteristic of the high-Andean zone, such as electrical discharges, which could generate interference or lead to circuit failure. Furthermore, the reliance on lithium-ion batteries will imply long-term operational costs due to the need for maintenance or replacement.

As part of future work, a more in-depth and detailed analysis of the system's energy consumption is proposed. Likewise, methods to shield the electronic circuits to protect them from electrical discharges during storms will be explored. The continued development of this technology will enable deployment into zones without internet connectivity, allowing for the collection of more multivariate information and, thus, a better understanding of environmental data behavior in high-Andean agricultural regions.

For the further enhancement of the prototype, a suggested future step is to utilize the collected historical data to apply artificial intelligence techniques. This would enable crucial predictions for farmer decision-making, especially in a zone that suffers from droughts, low temperatures, or shifts in planting dates. The implementation of artificial intelligence algorithms will allow for the full leveraging of this multivariate data for prediction in the agricultural sector, provided that the collection of reliable data is ensured to obtain equally reliable results when training the AI models.

Ethics in publishing statement

I testify on behalf of all co-authors that our article submitted followed ethical principles in publishing.

Title: LoRaWAN-based system for multivariate data acquisition in high Andean agriculture

All authors agree that:

This research presents an accurate account of the work performed, all data presented are accurate and methodologies detailed enough to permit others to replicate the work.

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All authors have been personally and actively involved in substantive work leading to the manuscript and will hold themselves jointly and individually responsible for its content.

CRediT authorship contribution statement

Godofredo Quea Lopez: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Angel Rosendo Condori Coaquira:** Supervision, Validation, Visualization, Writing – original draft. **Jorge Eddy Otazu Luque:** Investigation, Supervision, Writing – original draft. **Eder Gutierrez Quispe:** Conceptualization, Investigation, Supervision, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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